



7099 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Cycle: 4, Proposal Category: GO

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JWST Proposal 7099 (Created: Friday, July 11, 2025, 11:00:12AM Eastern Standard Time) - Overview

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OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
RhoOph				
	1	YLW19-M	MIRI Medium Resolution Spectroscopy	(1) YLW-19
	2	GSS29-M	MIRI Medium Resolution Spectroscopy	(2) GSS-29
	3	J162636-F	MIRI Medium Resolution Spectroscopy	(3) J162636.8-241552
	4	WL18-F	MIRI Medium Resolution Spectroscopy	(4) WL-18
	5	J162656-F	MIRI Medium Resolution Spectroscopy	(5) J162656.8-241351
	6	VSSG24-F	MIRI Medium Resolution Spectroscopy	(6) VSSG-24
	7	Eli233-F	MIRI Medium Resolution Spectroscopy	(7) Elia-2-33
	8	YLW18-B	MIRI Medium Resolution Spectroscopy	(8) YLW-18
	9	YLW47-M	MIRI Medium Resolution Spectroscopy	(9) YLW-47
	10	Sr10-F	MIRI Medium Resolution Spectroscopy	(10) SR-10
	11	YLW58-F	MIRI Medium Resolution Spectroscopy	(11) YLW-58

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
	12	Sr20W-F	MIRI Medium Resolution Spectroscopy	(12) SR-20W
	13	J162854-F	MIRI Medium Resolution Spectroscopy	(13) J162854.1-244744

ABSTRACT

Recent advances now allow us to statistically compare the compositions of planets with their planet-forming environments. Gas giant planets with masses within a factor of a few of Jupiter exhibit a wide range of atmospheric C/O ratios, including values greater than 1. A high C/O ratio has traditionally been interpreted as evidence that a planet formed beyond the water snowline and later migrated inward. However, recent JWST observations suggest that the C/O environment in protoplanetary disks is more complex and likely evolves over the first few million years (Myr) of the disk's lifetime. Specifically, recent JWST MIRI observations of young (< 1 Myr) Class II disks reveal chemistry consistent with high C/O ratios in the inner few astronomical units. This raises the possibility that such planets could form in situ, without significant migration. To better understand these environments, we propose a survey of 20 young Class II disks in the Rho Oph star-forming region, combining 7 archival datasets with 13 new JWST MIRI observations of star-disk systems. The sample is chosen based on having sensitive ALMA data enabling a comparison with the outer disk composition, including C/O. This expanded dataset will provide a more representative view of typical planet-forming environments around both M-dwarfs and solar-type stars, offering new insights into the evolution of C/O ratios in disks and their influence on the formation and composition of emerging planets.

OBSERVING DESCRIPTION

We propose to use JWST MIRI MRS to observe 13 disks targeted as part of the ALMA Disk-Exoplanet C/Onnection Large Program, that we will combine with 7 archival observations to get a complete sample of 20 young class II disks. Our lines of interest span the full MIRI range and thus we require full spectral scans of our sources from 5-28 micron. To detect the lines of interest requires SNR of 300 at 15 micron and >100 at 20 micron. Each observation begins with Target Acquisition and utilizes simultaneous imaging. Since we expect the inner disk mid-IR emission to be essentially point-like to MIRI, we adopt a four-point dither and do not carry out a dedicated background observation per source. Based on our ETC calculations, we adopt a FASTR1 readout pattern. The targets are grouped based on their known or expected mid-IR brightness (using Spitzer spectra when available), setting the exact ratio of groups and integrations to optimize for cosmic ray mitigation (faint sources) or saturation (bright sources). For the simultaneous imaging, we use the F560, F770W, and F1000W to see if we can serendipitously find any extended structures like nearby outflows or streamers given the relative youth of this SFR. We set our groups for each filter to 5, 8, and 10 respectively, and integrations to 4 for all three, which would allow us to see faint structures like what was reported in Mullin et al. 2024 as an additional qualitative signature of the relative youth of the region near each of our targets, while not saturating on a 12th K-magnitude K-dwarf star, thus allowing these data to also improve astrometric solutions if needed.

Proposal 7099 - Targets - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	YLW-19	RA: 16 25 36.7381 (246.4030754d) Dec: -24 15 42.53 (-24.26181d) Equinox: J2000	Proper Motion RA: -5.265 mas/yr Proper Motion Dec: -27.35599986866664 mas/yr Parallax: 0.006918300000000005" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks] Extended=NO				
	(2)	GSS-29	RA: 16 26 16.8562 (246.5702342d) Dec: -24 22 23.15 (-24.37310d) Equinox: J2000	Proper Motion RA: -6.920000000000001 mas/yr Proper Motion Dec: -27.2180000138178 mas/yr Parallax: 0.0072769" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks] Extended=NO				
	(3)	J162636.8-241552	RA: 16 26 36.8351 (246.6534796d) Dec: -24 15 51.85 (-24.26440d) Equinox: J2000	Proper Motion RA: -5.52 mas/yr Proper Motion Dec: -27.067999963037437 mas/yr Parallax: 0.0073212" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks] Extended=NO				
	(4)	WL-18	RA: 16 26 48.9810 (246.7040875d) Dec: -24 38 25.15 (-24.64032d) Equinox: J2000	Proper Motion RA: -4.845 mas/yr Proper Motion Dec: -25.295999944319192 mas/yr Parallax: 0.0075468" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks] Extended=NO					

Proposal 7099 - Targets - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

(5)	J162656.8-241351	RA: 16 26 56.7836 (246.7365983d) Dec: -24 13 51.54 (-24.23098d) Equinox: J2000	Proper Motion RA: -10.047 mas/yr Proper Motion Dec: -25.20600007756002 mas/yr Parallax: 0.0073251" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Protoplanetary disks] Extended=NO			
(6)	VSSG-24	RA: 16 27 13.7329 (246.8072204d) Dec: -24 18 16.86 (-24.30468d) Equinox: J2000	Proper Motion RA: -7.272 mas/yr Proper Motion Dec: -27.347999935045664 mas/yr Parallax: 0.0072888" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Protoplanetary disks] Extended=NO			
(7)	Elia-2-33	RA: 16 27 30.1802 (246.8757508d) Dec: -24 27 43.37 (-24.46205d) Equinox: J2000	Proper Motion RA: -7 mas/yr Proper Motion Dec: -25 mas/yr Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Proper motion estimated as the average value from the rest of the stars in the cluster.</i> Category=Star Description=[Protoplanetary disks] Extended=NO			
(8)	YLW-18	RA: 16 27 33.1158 (246.8879825d) Dec: -24 41 15.26 (-24.68757d) Equinox: J2000	Proper Motion RA: -8.723 mas/yr Proper Motion Dec: -25.805000018408464 mas/yr Parallax: 0.0073235" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Protoplanetary disks] Extended=NO			
(9)	YLW-47	RA: 16 27 38.3206 (246.9096692d) Dec: -24 36 58.59 (-24.61627d) Equinox: J2000	Proper Motion RA: -6.268 mas/yr Proper Motion Dec: -25.180000011459924 mas/yr Parallax: 0.0071705" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Protoplanetary disks] Extended=NO			

Proposal 7099 - Targets - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

(10)	SR-10	RA: 16 27 55.5811 (246.9815879d) Dec: -24 26 18.03 (-24.43834d) Equinox: J2000	Proper Motion RA: -9.112 mas/yr Proper Motion Dec: -26.40500001689361 mas/yr Parallax: 0.0075961" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks] Extended=NO			
(11)	YLW-58	RA: 16 28 16.5129 (247.0688038d) Dec: -24 36 58.05 (-24.61613d) Equinox: J2000	Proper Motion RA: -6.928 mas/yr Proper Motion Dec: -26.02200004275801 mas/yr Parallax: 0.007408599999999996" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks] Extended=NO			
(12)	SR-20W	RA: 16 28 23.3449 (247.0972704d) Dec: -24 22 40.67 (-24.37796d) Equinox: J2000	Proper Motion RA: -6.321 mas/yr Proper Motion Dec: -23.964999968484335 mas/yr Parallax: 0.0068116" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks] Extended=NO			
(13)	J162854.1-244744	RA: 16 28 54.0781 (247.2253254d) Dec: -24 47 44.29 (-24.79564d) Equinox: J2000	Proper Motion RA: -7.159 mas/yr Proper Motion Dec: -25.376000053256575 mas/yr Parallax: 0.0073279" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks] Extended=NO			

Proposal 7099 - Observation 1 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Observation	Proposal 7099, Observation 1: YLW19-M													Fri Jul 11 16:00:12 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(1)	YLW-19	RA: 16 25 36.7381 (246.4030754d) Dec: -24 15 42.53 (-24.26181d) Equinox: J2000				Proper Motion RA: -5.265 mas/yr Proper Motion Dec: -27.355999986866664 mas/yr Parallax: 0.0069183000000000005" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SAME	FND	FAST	6	1		1	16.65	225449				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	225449	
	1	LONG(C)	MRSLONG		FASTR1	15	3	1	Dither 1	4	12	521.708	225449	
	1	LONG(C)	MRSSHORT		FASTR1	15	3	1	Dither 1	4	12	521.708	225449	
	2		IMAGER	F770W	FASTR1	8	4	1	Dither 1	4	16	388.506	225449	
	2	MEDIUM(B)	MRSLONG		FASTR1	11	3	1	Dither 1	4	12	388.506	225449	
	2	MEDIUM(B)	MRSSHORT		FASTR1	11	3	1	Dither 1	4	12	388.506	225449	
	3		IMAGER	F560W	FASTR1	5	4	1	Dither 1	4	16	255.304	225449	
	3	SHORT(A)	MRSLONG		FASTR1	12	3	1	Dither 1	4	12	421.806	225449	
	3	SHORT(A)	MRSSHORT		FASTR1	12	3	1	Dither 1	4	12	421.806	225449	

Proposal 7099 - Observation 2 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Observation	Proposal 7099, Observation 2: GSS29-M													Fri Jul 11 16:00:12 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(2)	GSS-29	RA: 16 26 16.8562 (246.5702342d) Dec: -24 22 23.15 (-24.37310d) Equinox: J2000				Proper Motion RA: -6.920000000000001 mas/yr Proper Motion Dec: -27.2180000138178 mas/yr Parallax: 0.0072769" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID					
	1	SAME	FND	FAST	4	1	1	11.1	225449					
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	225449	
	1	LONG(C)	MRSLONG		FASTR1	15	3	1	Dither 1	4	12	521.708	225449	
	1	LONG(C)	MRSSHORT		FASTR1	15	3	1	Dither 1	4	12	521.708	225449	
	2		IMAGER	F770W	FASTR1	8	4	1	Dither 1	4	16	388.506	225449	
	2	MEDIUM(B)	MRSLONG		FASTR1	11	3	1	Dither 1	4	12	388.506	225449	
	2	MEDIUM(B)	MRSSHORT		FASTR1	11	3	1	Dither 1	4	12	388.506	225449	
	3		IMAGER	F560W	FASTR1	5	4	1	Dither 1	4	16	255.304	225449	
	3	SHORT(A)	MRSLONG		FASTR1	12	3	1	Dither 1	4	12	421.806	225449	
	3	SHORT(A)	MRSSHORT		FASTR1	12	3	1	Dither 1	4	12	421.806	225449	

Proposal 7099 - Observation 3 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Observation	Proposal 7099, Observation 3: J162636-F													Fri Jul 11 16:00:12 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(3)	J162636.8-241552	RA: 16 26 36.8351 (246.6534796d) Dec: -24 15 51.85 (-24.26440d) Equinox: J2000				Proper Motion RA: -5.52 mas/yr Proper Motion Dec: -27.067999963037437 mas/yr Parallax: 0.0073212" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID					
	1	SAME	FND	FAST	4	1	1	11.1	225449					
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	225449	
	1	LONG(C)	MRSLONG		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	1	LONG(C)	MRSSHORT		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	2		IMAGER	F770W	FASTR1	8	4	1	Dither 1	4	16	388.506	225449	
	2	MEDIUM(B)	MRSLONG		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	2	MEDIUM(B)	MRSSHORT		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	3		IMAGER	F560W	FASTR1	5	4	1	Dither 1	4	16	255.304	225449	
	3	SHORT(A)	MRSLONG		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	
	3	SHORT(A)	MRSSHORT		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	

Proposal 7099 - Observation 4 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Observation	Proposal 7099, Observation 4: WL18-F												Fri Jul 11 16:00:12 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(4)	WL-18	RA: 16 26 48.9810 (246.7040875d) Dec: -24 38 25.15 (-24.64032d) Equinox: J2000				Proper Motion RA: -4.845 mas/yr Proper Motion Dec: -25.295999944319192 mas/yr Parallax: 0.0075468" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Protoplanetary disks] Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SAME	FND	FAST	4	1	1	11.1	225449				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		YES			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	225449
	1	LONG(C)	MRSLONG		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449
	1	LONG(C)	MRSSHORT		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449
	2		IMAGER	F770W	FASTR1	8	4	1	Dither 1	4	16	388.506	225449
	2	MEDIUM(B)	MRSLONG		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449
	2	MEDIUM(B)	MRSSHORT		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449
	3		IMAGER	F560W	FASTR1	5	4	1	Dither 1	4	16	255.304	225449
	3	SHORT(A)	MRSLONG		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449
	3	SHORT(A)	MRSSHORT		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449

Proposal 7099 - Observation 5 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Observation	Proposal 7099, Observation 5: J162656-F													Fri Jul 11 16:00:12 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(5)	J162656.8-241351	RA: 16 26 56.7836 (246.7365983d) Dec: -24 13 51.54 (-24.23098d) Equinox: J2000				Proper Motion RA: -10.047 mas/yr Proper Motion Dec: -25.20600007756002 mas/yr Parallax: 0.0073251" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SAME	FND	FAST	4	1		1	11.1	225449				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray				Grating Wheel Direction				
	All MRS		YES			FULL				Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	225449	
	1	LONG(C)	MRSLONG		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	1	LONG(C)	MRSSHORT		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	2		IMAGER	F770W	FASTR1	8	4	1	Dither 1	4	16	388.506	225449	
	2	MEDIUM(B)	MRSLONG		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	2	MEDIUM(B)	MRSSHORT		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	3		IMAGER	F560W	FASTR1	5	4	1	Dither 1	4	16	255.304	225449	
	3	SHORT(A)	MRSLONG		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	
	3	SHORT(A)	MRSSHORT		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	

Proposal 7099 - Observation 6 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Observation	Proposal 7099, Observation 6: VSSG24-F												Fri Jul 11 16:00:12 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(6)	VSSG-24	RA: 16 27 13.7329 (246.8072204d) Dec: -24 18 16.86 (-24.30468d) Equinox: J2000				Proper Motion RA: -7.272 mas/yr Proper Motion Dec: -27.347999935045664 mas/yr Parallax: 0.0072888" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID				
	1	SAME	FND	FAST	4	1		1	11.1	225449				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	225449	
	1	LONG(C)	MRSLONG		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	1	LONG(C)	MRSSHORT		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	2		IMAGER	F770W	FASTR1	8	4	1	Dither 1	4	16	388.506	225449	
	2	MEDIUM(B)	MRSLONG		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	2	MEDIUM(B)	MRSSHORT		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	3		IMAGER	F560W	FASTR1	5	4	1	Dither 1	4	16	255.304	225449	
	3	SHORT(A)	MRSLONG		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	
	3	SHORT(A)	MRSSHORT		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	

Proposal 7099 - Observation 7 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Observation	Proposal 7099, Observation 7: Eli233-F													Fri Jul 11 16:00:12 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(7)	Elia-2-33	RA: 16 27 30.1802 (246.8757508d)				Proper Motion RA: -7 mas/yr							
			Dec: -24 27 43.37 (-24.46205d)				Proper Motion Dec: -25 mas/yr							
			Equinox: J2000				Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	Proper motion estimated as the average value from the rest of the stars in the cluster. Category=Star Description=[Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations		Total Exposure Time		ETC Wkbk.Calc ID		
	1	SAME	FND	FAST	4	1		1		11.1		225449		
Template	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS			YES			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	225449	
	1	LONG(C)	MRSLONG		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	1	LONG(C)	MRSSHORT		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	2		IMAGER	F770W	FASTR1	8	4	1	Dither 1	4	16	388.506	225449	
	2	MEDIUM(B)	MRSLONG		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	2	MEDIUM(B)	MRSSHORT		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	3		IMAGER	F560W	FASTR1	5	4	1	Dither 1	4	16	255.304	225449	
	3	SHORT(A)	MRSLONG		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	
	3	SHORT(A)	MRSSHORT		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	

Proposal 7099 - Observation 8 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Observation	Proposal 7099, Observation 8: YLW18-B													Fri Jul 11 16:00:12 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(8)	YLW-18	RA: 16 27 33.1158 (246.8879825d) Dec: -24 41 15.26 (-24.68757d) Equinox: J2000				Proper Motion RA: -8.723 mas/yr Proper Motion Dec: -25.805000018408464 mas/yr Parallax: 0.0073235" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID					
	1	SAME	FND	FAST	6	1	1	16.65	225595					
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	225595	
	1	LONG(C)	MRSLONG		FASTR1	18	5	1	Dither 1	4	20	1043.415	225595	
	1	LONG(C)	MRSSHORT		FASTR1	18	5	1	Dither 1	4	20	1043.415	225595	
	2		IMAGER	F770W	FASTR1	8	4	1	Dither 1	4	16	388.506	225595	
	2	MEDIUM(B)	MRSLONG		FASTR1	19	5	1	Dither 1	4	20	1098.916	225595	
	2	MEDIUM(B)	MRSSHORT		FASTR1	19	5	1	Dither 1	4	20	1098.916	225595	
	3		IMAGER	F560W	FASTR1	5	4	1	Dither 1	4	16	255.304	225595	
	3	SHORT(A)	MRSLONG		FASTR1	22	5	1	Dither 1	4	20	1265.418	225595	
	3	SHORT(A)	MRSSHORT		FASTR1	22	5	1	Dither 1	4	20	1265.418	225595	

Proposal 7099 - Observation 9 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Observation	Proposal 7099, Observation 9: YLW47-M													Fri Jul 11 16:00:12 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(9)	YLW-47	RA: 16 27 38.3206 (246.9096692d) Dec: -24 36 58.59 (-24.61627d) Equinox: J2000				Proper Motion RA: -6.268 mas/yr Proper Motion Dec: -25.180000011459924 mas/yr Parallax: 0.0071705" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID					
	1	SAME	FND	FAST	6	1	1	16.65	225449					
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	225449	
	1	LONG(C)	MRSLONG		FASTR1	15	3	1	Dither 1	4	12	521.708	225449	
	1	LONG(C)	MRSSHORT		FASTR1	15	3	1	Dither 1	4	12	521.708	225449	
	2		IMAGER	F770W	FASTR1	8	4	1	Dither 1	4	16	388.506	225449	
	2	MEDIUM(B)	MRSLONG		FASTR1	11	3	1	Dither 1	4	12	388.506	225449	
	2	MEDIUM(B)	MRSSHORT		FASTR1	11	3	1	Dither 1	4	12	388.506	225449	
	3		IMAGER	F560W	FASTR1	5	4	1	Dither 1	4	16	255.304	225449	
	3	SHORT(A)	MRSLONG		FASTR1	12	3	1	Dither 1	4	12	421.806	225449	
	3	SHORT(A)	MRSSHORT		FASTR1	12	3	1	Dither 1	4	12	421.806	225449	

Proposal 7099 - Observation 10 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Observation	Proposal 7099, Observation 10: Sr10-F													Fri Jul 11 16:00:12 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(10)	SR-10	RA: 16 27 55.5811 (246.9815879d) Dec: -24 26 18.03 (-24.43834d) Equinox: J2000				Proper Motion RA: -9.112 mas/yr Proper Motion Dec: -26.40500001689361 mas/yr Parallax: 0.0075961" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID					
	1	SAME	FND	FAST	6	1	1	16.65	225449					
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction					
	All MRS		YES			FULL			Allow Auto Reorder					
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	225449	
	1	LONG(C)	MRSLONG		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	1	LONG(C)	MRSSHORT		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	2		IMAGER	F770W	FASTR1	8	4	1	Dither 1	4	16	388.506	225449	
	2	MEDIUM(B)	MRSLONG		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	2	MEDIUM(B)	MRSSHORT		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	3		IMAGER	F560W	FASTR1	5	4	1	Dither 1	4	16	255.304	225449	
	3	SHORT(A)	MRSLONG		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	
	3	SHORT(A)	MRSSHORT		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	

Proposal 7099 - Observation 11 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Observation	Proposal 7099, Observation 11: YLW58-F													Fri Jul 11 16:00:12 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(11)	YLW-58	RA: 16 28 16.5129 (247.0688038d) Dec: -24 36 58.05 (-24.61613d) Equinox: J2000				Proper Motion RA: -6.928 mas/yr Proper Motion Dec: -26.02200004275801 mas/yr Parallax: 0.007408599999999996" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID					
	1	SAME	FND	FAST	6	1	1	16.65	225449					
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	225449	
	1	LONG(C)	MRSLONG		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	1	LONG(C)	MRSSHORT		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	2		IMAGER	F770W	FASTR1	8	4	1	Dither 1	4	16	388.506	225449	
	2	MEDIUM(B)	MRSLONG		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	2	MEDIUM(B)	MRSSHORT		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	3		IMAGER	F560W	FASTR1	5	4	1	Dither 1	4	16	255.304	225449	
	3	SHORT(A)	MRSLONG		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	
	3	SHORT(A)	MRSSHORT		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	

Proposal 7099 - Observation 12 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Observation	Proposal 7099, Observation 12: Sr20W-F													Fri Jul 11 16:00:12 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(12)	SR-20W	RA: 16 28 23.3449 (247.0972704d) Dec: -24 22 40.67 (-24.37796d) Equinox: J2000				Proper Motion RA: -6.321 mas/yr Proper Motion Dec: -23.964999968484335 mas/yr Parallax: 0.0068116" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID					
	1	SAME	FND	FAST	6	1	1	16.65	225449					
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction					
	All MRS		YES			FULL			Allow Auto Reorder					
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	225449	
	1	LONG(C)	MRSLONG		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	1	LONG(C)	MRSSHORT		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	2		IMAGER	F770W	FASTR1	8	4	1	Dither 1	4	16	388.506	225449	
	2	MEDIUM(B)	MRSLONG		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	2	MEDIUM(B)	MRSSHORT		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	3		IMAGER	F560W	FASTR1	5	4	1	Dither 1	4	16	255.304	225449	
	3	SHORT(A)	MRSLONG		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	
	3	SHORT(A)	MRSSHORT		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	

Proposal 7099 - Observation 13 - Unveiling the Changing C/O Chemistry in Young Planet-Forming Disks with JWST and ALMA

Observation	Proposal 7099, Observation 13: J162854-F													Fri Jul 11 16:00:13 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(13)	J162854.1-244744	RA: 16 28 54.0781 (247.2253254d) Dec: -24 47 44.29 (-24.79564d) Equinox: J2000				Proper Motion RA: -7.159 mas/yr Proper Motion Dec: -25.376000053256575 mas/yr Parallax: 0.0073279" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Protoplanetary disks] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID					
	1	SAME	FND	FAST	6	1	1	16.65	225449					
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction					
	All MRS		YES			FULL			Allow Auto Reorder					
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	10	4	1	Dither 1	4	16	477.307	225449	
	1	LONG(C)	MRSLONG		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	1	LONG(C)	MRSSHORT		FASTR1	25	5	1	Dither 1	4	20	1431.921	225449	
	2		IMAGER	F770W	FASTR1	8	4	1	Dither 1	4	16	388.506	225449	
	2	MEDIUM(B)	MRSLONG		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	2	MEDIUM(B)	MRSSHORT		FASTR1	25	4	1	Dither 1	4	16	1143.316	225449	
	3		IMAGER	F560W	FASTR1	5	4	1	Dither 1	4	16	255.304	225449	
	3	SHORT(A)	MRSLONG		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	
	3	SHORT(A)	MRSSHORT		FASTR1	31	4	1	Dither 1	4	16	1409.72	225449	